$$Ab_2 = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} (-1)\cdot(-2)+2\cdot 1 \\ 5\cdot(-2)+4\cdot 1 \\ 2\cdot(-2)+(-3)\cdot 1 \end{bmatrix} \text{ (Row-Vector Rule)}$$

$$= \begin{bmatrix} 4 \\ -6 \\ -7 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} -7 & 4 \\ 7 & -6 \\ 12 & -7 \end{bmatrix}$$

b) $$AB = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} (-1)\cdot 3+2\cdot(-2) & (-1)\cdot(-2)+2\cdot 1 \\ 5\cdot 3+4\cdot(-2) & 5\cdot(-2)+4\cdot 1 \\ 2\cdot 3+(-3)\cdot(-2) & 2\cdot(-2)+(-3)\cdot 1 \end{bmatrix}$$

(Row-Column Rule for Computing AB)

$$= \begin{bmatrix} -7 & 4 \\ 7 & -6 \\ 12 & -7 \end{bmatrix}$$

7. If a matrix A is 5×3 and the product AB is 5×7, what is the size of B?

Ans: By the defn. of matrix multiplication, since AB is a valid product, the no. of columns of A and the no. of rows of B must be the same. $\therefore$ B has 3 rows. The no. of columns of AB must be the same as that of B. $\therefore$ B has 7 columns, since AB has 7 columns.

$\boxed{\therefore B \text{ is a } 3\times 7 \text{ matrix}}$

9. Let $A = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -5 \\ 3 & k \end{bmatrix}$. What values of k, if any, will make $AB = BA$?

Solution: $AB = [Ab_1 \; Ab_2]$ (By defn. of matrix multiplication)

$$Ab_1 = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 4\begin{bmatrix} 2 \\ -3 \end{bmatrix} + 3\begin{bmatrix} 5 \\ 1 \end{bmatrix} \text{ (By defn. of } Ax\text{)}$$

$$= \begin{bmatrix} 8 \\ -12 \end{bmatrix} + \begin{bmatrix} 15 \\ 3 \end{bmatrix} \text{ (By defn. of vector scalar multiple)}$$

$$= \begin{bmatrix} 23 \\ -9 \end{bmatrix} \text{ (By defn. of vector sum)}$$